

Histology Without the Blade: Can We Virtually Stain Thick Tissues with Pathological Accuracy?

Jintaek Oh, Lulin Shi, Terence T. W. Wong Translational and Advanced Bioimaging Laboratory, The Hong Kong University of Science and Technology  ttwwong@ust.hk

THE PROBLEM

Gold-standard method for diagnosing diseases from tissue samples is slow and laborious.

Existing "virtual staining" methods are a step forward, but they often:

- Still require the slow tissue sectioning step.
- Fail to capture complex, clinically-relevant cancer features.

Fixation → Dehydration → Embedding → Sectioning → Mounting → staining



QUANTITATIVE EVALUATION

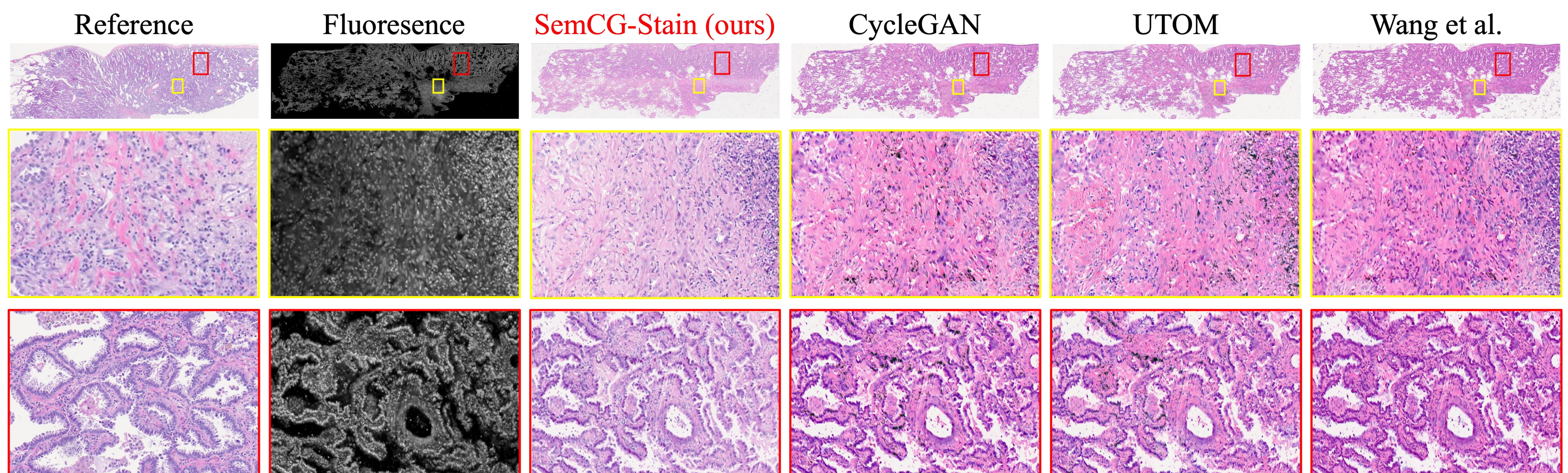
SemCG-Stain outperforms SOTA methods across all standard metrics, confirming higher image fidelity.

	FID (↓)	KID (x100) (↓)	IS (↑)
Cycle GAN ¹	76.51	6.05 ± 0.34	3.02 ± 0.07
UTOM ²	<u>64.11</u>	4.84 ± 0.24	3.05 ± 0.04
Wang et al. ³	64.67	<u>4.44 ± 0.21</u>	<u>3.28 ± 0.06</u>
SemCG-Stain	54.57	2.47 ± 0.13	3.39 ± 0.06

- **Bold** entries = best (lowest FID/KID, highest IS).
- Underlined entries = second best.

QUALITATIVE EVALUATION

Our method produces cleaner and more natural H&E stains that preserve fine details.



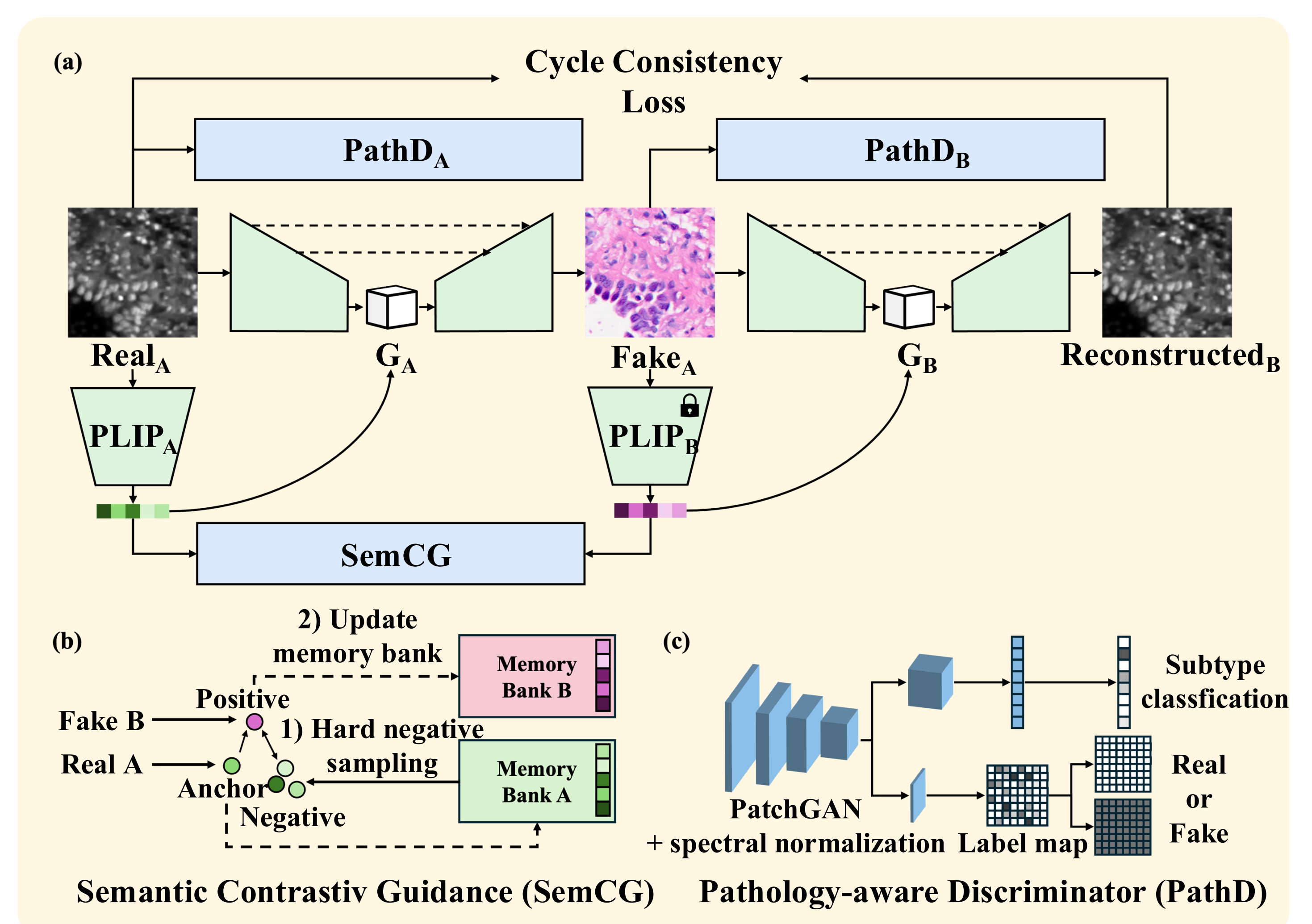
WHAT WE DID

Developed an AI that generates fast and accurate virtual H&E stains directly from thick, unprocessed tissue.

1. Semantic Contrastive Guidance (SemCG)
 - Morphological consistency.
 - Features from pathology foundation model.
2. Pathology-aware Discriminator (PathD)
 - Teach the discriminator to think like a pathologist.
 - Real vs. Fake AND classify the Cancer Subtype.

WHAT'S NEXT

- Extend to more tissue types and diseases.
- Explore other section-free imaging modalities.
- Conduct clinical validation studies for diagnostic accuracy.



Acknowledgement

This work was supported in part by the Hong Kong Innovation and Technology Commission (ITS/023/22FP).

References

1. Zhu, J.Y., Park, T., Isola, P., Efros, A.A.: Unpaired image-to-image translation using cycle-consistent adversarial networks. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2223–2232 (2017)
2. Li, X., et al.: Unsupervised content-preserving transformation for optical microscopy. Light: Sci. Appl. 10(1), 1–11 (2021)
3. Wang, R., et al.: Virtual bright field and fluorescence staining for fourier ptychography via unsupervised deep learning. Opt. Lett. 45 (19), 5405–5408 (2020)

